

Hart Luo 骆皓天

Frontend Engineer · UNSW BSc & MIT · Focus: *canvas interaction and complex front-end state*

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00 · ABOUT

Highlights

- 01 Frontend engineer on Hithink RoyalFlush's DreamFace, an overseas AIGC product with **300k+** DAU; helped take its infinite canvas from zero to launch, owning the document protocol, save pipeline and render-state consistency.
- 02 Author of the open-source canvas engine infinite-canvas-core, optimised against benchmarks: **48 μs** viewport queries over 100,000 nodes, worst-case frame 44 → **13.8 ms**.
- 03 BSc in Computer Science and Master of IT from UNSW (GPA 6.03 and 6.12 out of 7); nearly five years studying and interning in Australia; works fluently in English.

01 · WORK

Experience

2026.04

Present

Hithink RoyalFlush Frontend Engineer

DreamFace team · overseas AIGC product, **300k+ DAU** on iOS, Android and web (talking avatars, image-to-video, motion transfer, voice cloning); I work on the web app, mobile web and admin console

[Vue 3](#) / [TypeScript](#) / [Canvas](#) / [Analytics](#)

Infinite canvas · AIGC creative workspace core developer · zero to launch · 3,000+ DAU

- Ran a **26**-point feasibility study of the Woven Canvas engine that informed project approval; built the canvas route, workspace and resizable agent panel from scratch.
- Designed the protocol layer — **CanvasDocument** model, incremental command protocol, undo / redo history and backend spec (**4** design docs) — and moved saving from localStorage to the server.
- Built the incremental save queue: debounced merging, idempotent retries reusing the request ID, revision-conflict retry, and weighted batching that fixed failed saves on **300**-node bulk edits.
- Fixed **5** root causes of state divergence between renderer and document (ghost nodes, selection snap-back, a render-loop crash on delete, offline freezes, bulk-asset render failures), with regression tests.
- Implemented zoom steps, viewport anchoring, snapping, marquee selection, resize, mirroring, drag-and-drop upload and Safari fixes; presented the canvas architecture to the team.

Promotions and analytics

- Shipped the site-wide top banner and the image / video campaign pop-up end to end (front end and admin console); owned event tracking for core modules, catching missing triggers and conflicting IDs before release.

02 · INTERN

Internships

2025.06

2025.11

DVINE Experiences Software / IT Intern · Australian tourism company

Booking and operations automation on the GoHighLevel low-code platform

[GoHighLevel](#) / [Forms](#) / [Workflows](#)

- Built configuration-driven booking forms on GHL Forms / Custom Fields and trigger → action workflows covering booking creation, confirmations, status updates and reminders; shipped **10+** forms and modules.

2024.06

2024.09

Beyondsoft Frontend Intern · on site with an Alibaba project team

Supply-chain admin system (enterprise SaaS) delivered for ChaPanda; owned the transport management

and order visualisation modules

[React](#) / [Umi](#) / [Ant Design Pro](#) / [ECharts](#)

— Shipped **10+** pages with Umi, React, TypeScript and Ant Design Pro, **10+** reusable components and **20+** ECharts reports, from requirements through release.

03 · BUILD

Projects

2026.09 Open source	infinite-canvas-core Headless infinite-canvas engine github ↗ General canvas capabilities rebuilt in my own time as a standalone open-source engine (no company code) TypeScript / Canvas2D / Vitest / Playwright — Spatial index : a uniform grid with cell coordinates packed exactly into one float64 and oversized items kept apart — 48 μs viewport queries over 100,000 nodes (604 μs for a linear scan, 13×). — Rendering : batching and a sub-pixel LOD buffer cut the worst 100k-node frame from 44 to 13.8 ms ; benchmarks caught a 3.3× regression from downlevelled <code>#private</code> fields, now test-guarded. — Architecture : scale-and-translate camera, geometric hit testing, patch-based undo and snapping; DOM-free core runs in a Worker or Node; 160+ unit tests.
2026.08 Open source	Hand Review Trainer Offline-first PWA live ↗ A full training loop: play → record → EV analysis → flag mistakes → review → long-term statistics React / TypeScript / IndexedDB / Playwright — Rules engine, 6 AI strategies, per-decision EV review and 15 mistake categories; state is separated from UI and all randomness comes from a seed, so any session replays exactly. — 985 automated tests: 100,000 cases cross-checked against a reference, invariants over 10,000 self-play sessions, Playwright on desktop and mobile; works offline.
At UNSW Startup	UNSW HandbookX Frontend lead · course search and planning github ↗ Built because the official course handbook was slow to use; 100+ student users during campus testing Vue 3 / TypeScript / NestJS — Set the team's coding conventions and component style guide, ran weekly code reviews, and built shared components and utilities to cut duplicated work.
More Work	Nimbus Scheduler, a distributed task scheduler (Node.js / WebSocket / Vue 3); RepoQuest, a pixel-map explorer for GitHub repositories (React / TypeScript); and more at haotian14.github.io/Portfolio .

04 · STUDY

Education

2024.09 2026.01	UNSW Sydney Master of Information Technology · GPA 6.12 / 7.0 · QS 19
2021.09 2024.06	UNSW Sydney Bachelor of Science (Computer Science) · GPA 6.03 / 7.0 · QS 19

05 · CRAFT

Skills

FRONTEND	Proficient in Vue 3 , TypeScript and JavaScript; experienced with React , Umi, Vite, Ant Design, TailwindCSS, Zustand, Sass, ECharts, PWA / IndexedDB
GRAPHICS	Canvas2D and complex interaction: coordinate transforms, spatial indexing, viewport culling, incremental rendering; document models, incremental saving and undo / redo design
ENGINEERING	Vitest / Playwright testing and regression suites, GitHub Actions CI, Sonar; benchmarking and profiling, analytics design; Claude Code / Codex for day-to-day development and code review
OTHER	Node.js, WebSocket, RESTful API design; data structures and algorithms
LANGUAGE	Mandarin (native), English (professional working proficiency)